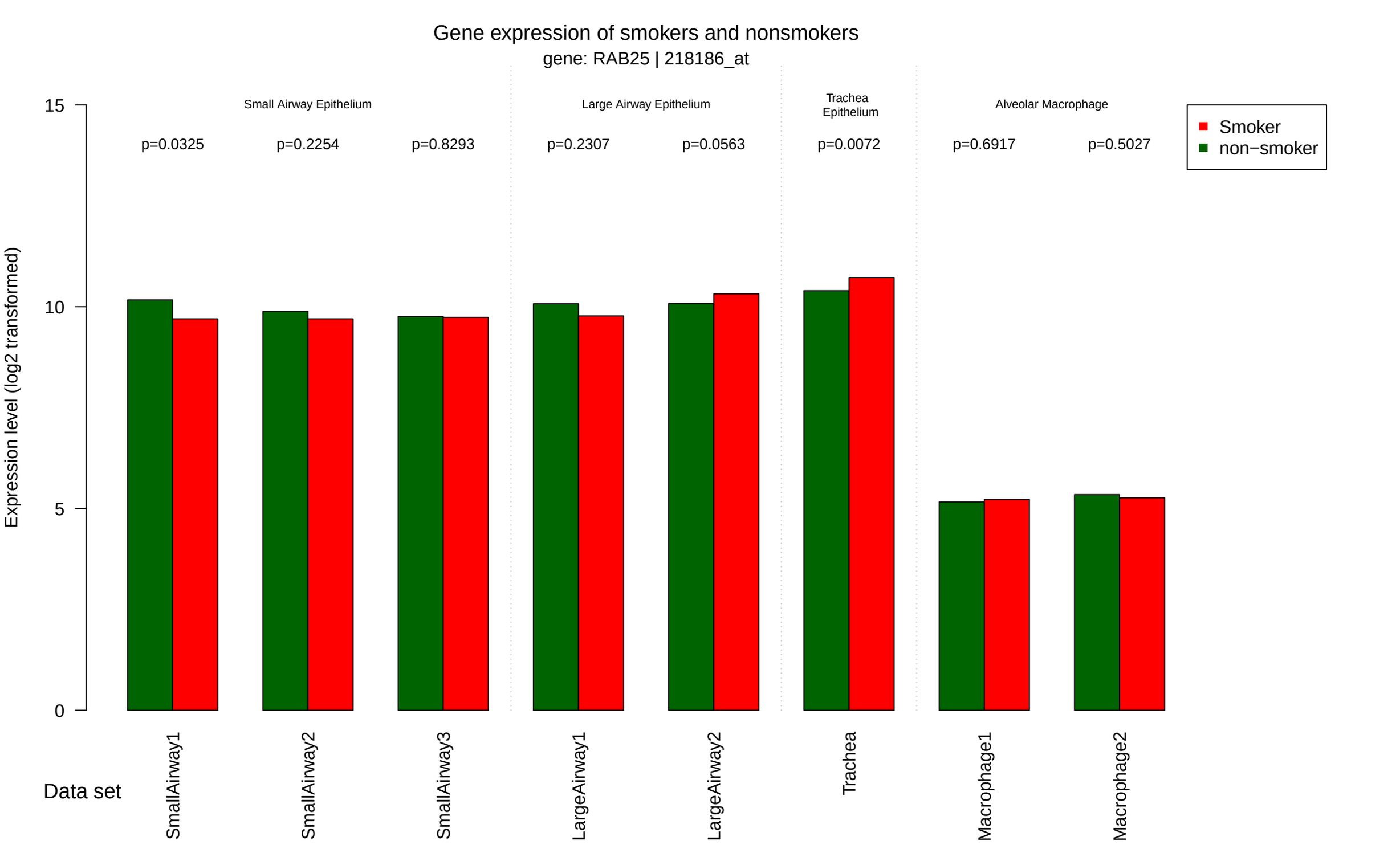
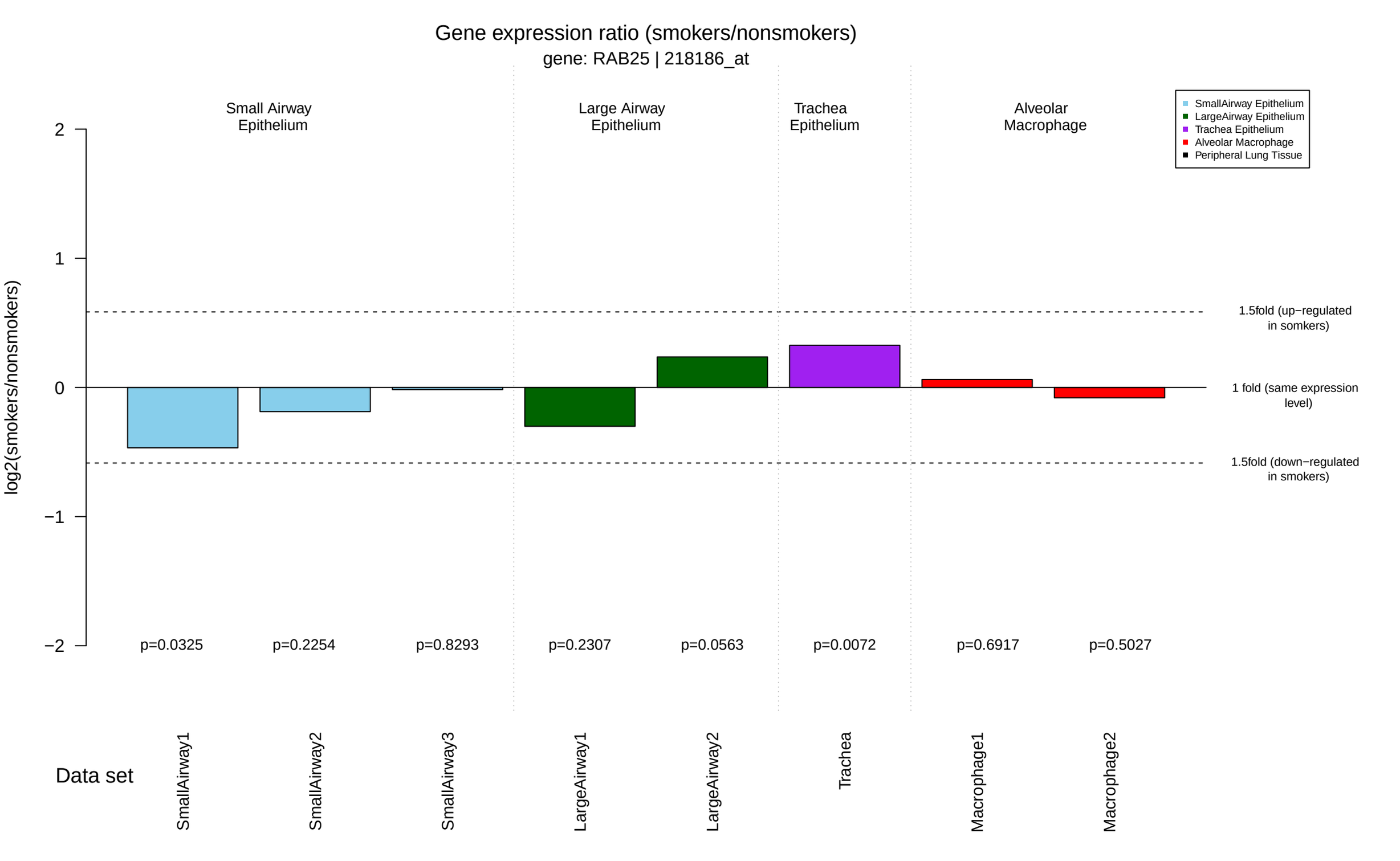
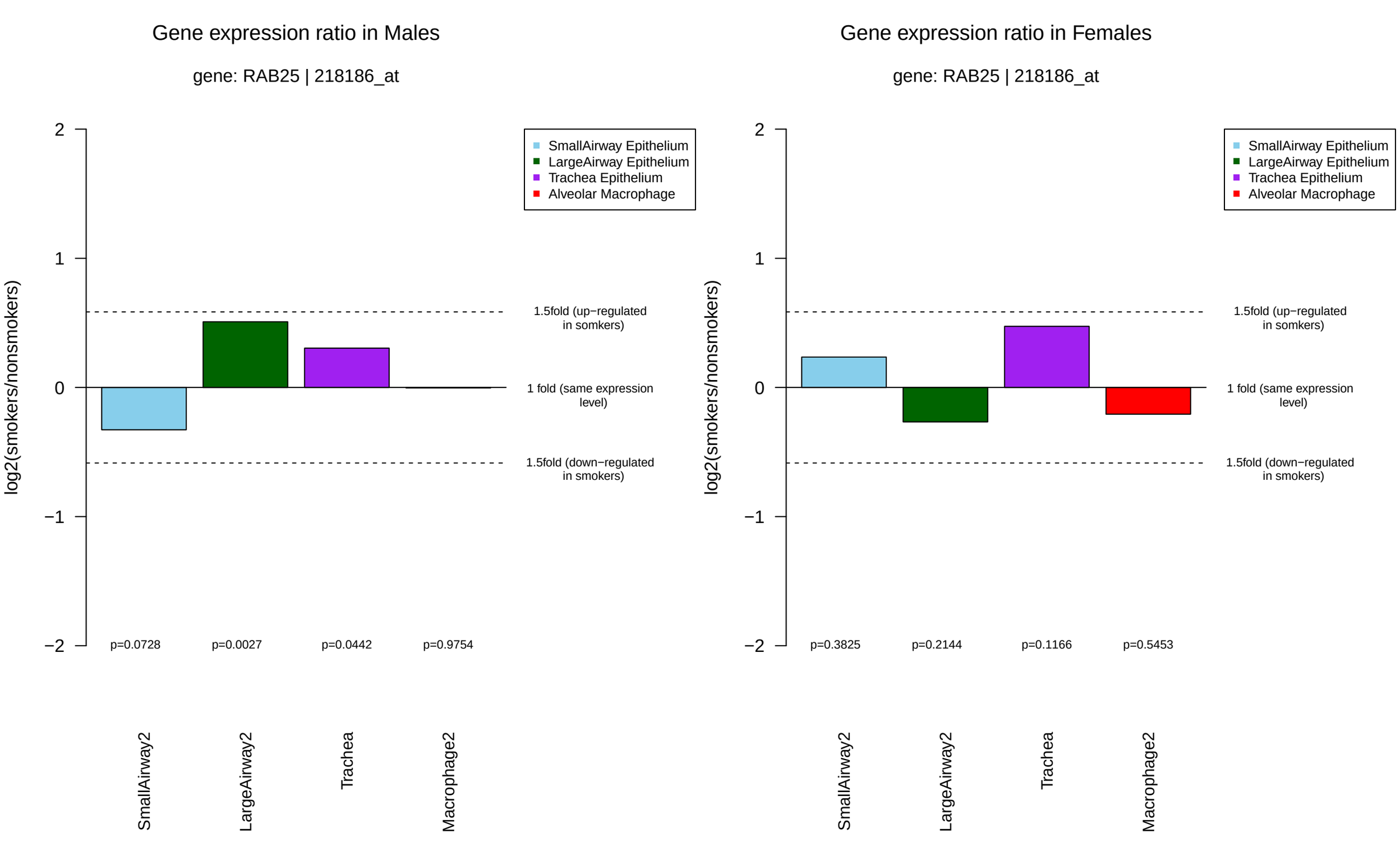
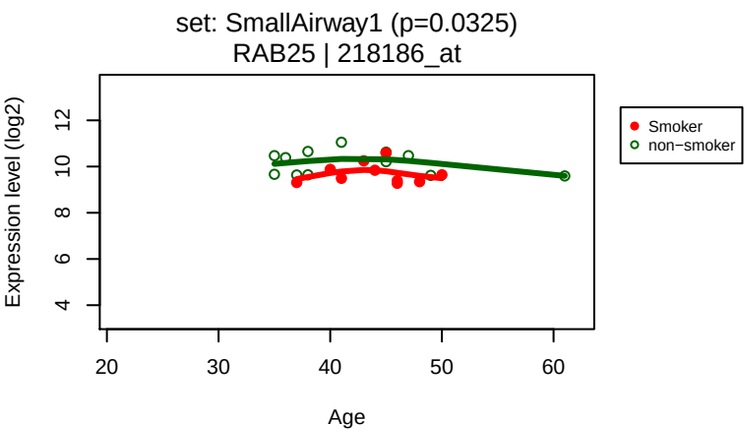
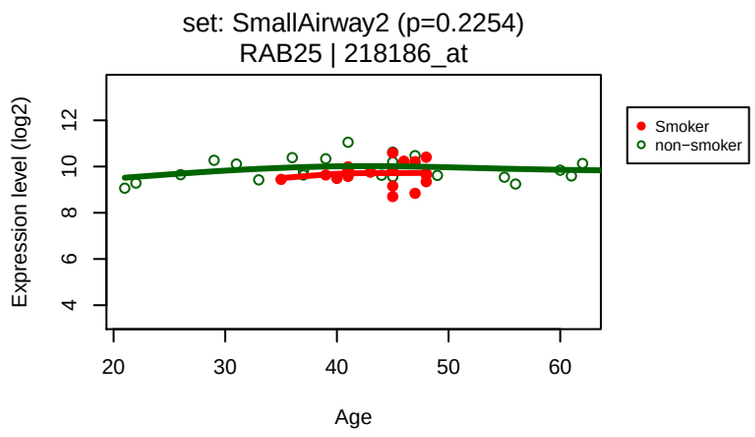
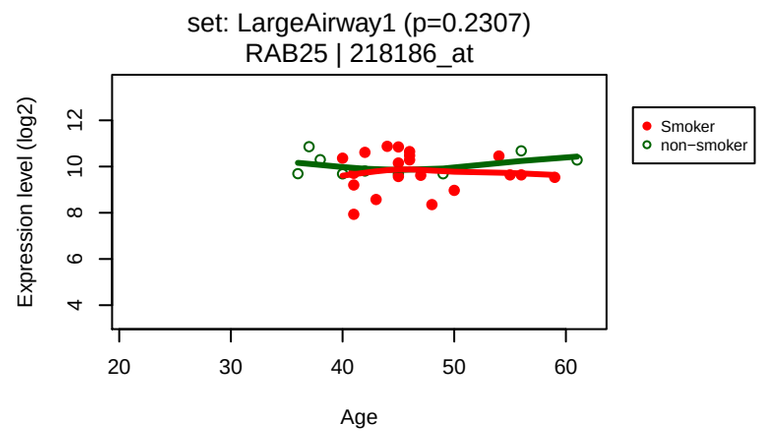

Gene expression of smokers and nonsmokers
gene: RAB25 | 218186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0325
p=0.2254
p=0.8293
p=0.2307
p=0.0563
p=0.0072
p=0.6917
p=0.5027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

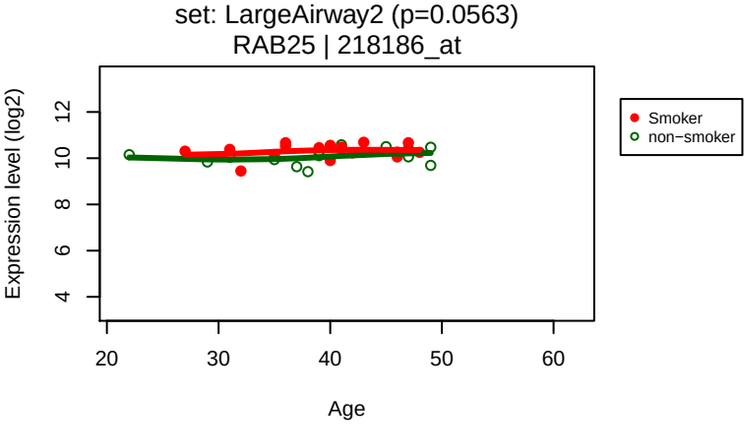

Gene expression ratio (smokers/nonsmokers)
gene: RAB25 | 218186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.2254
p=0.8293
p=0.2307
p=0.0563
p=0.0072
p=0.6917
p=0.5027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

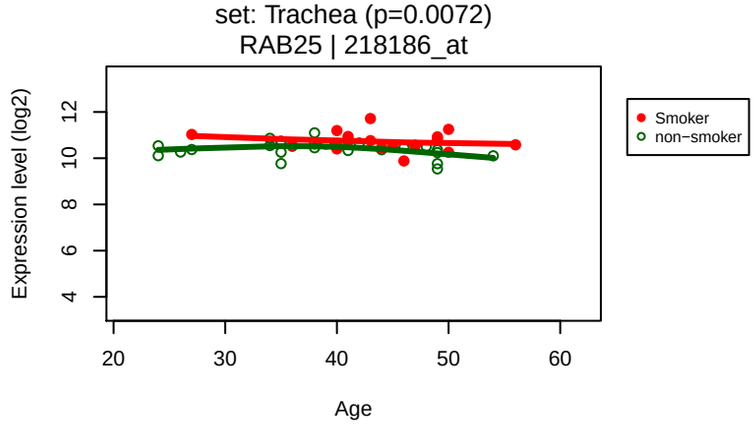

Gene expression ratio in Males
gene: RAB25 | 218186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0728
p=0.0027
p=0.0442
p=0.9754
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB25 | 218186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3825
p=0.2144
p=0.1166
p=0.5453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

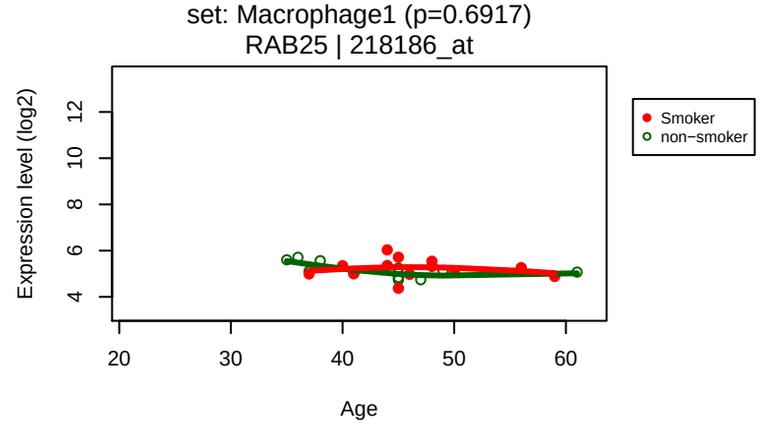

set: SmallAirway1 (p=0.0325)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker

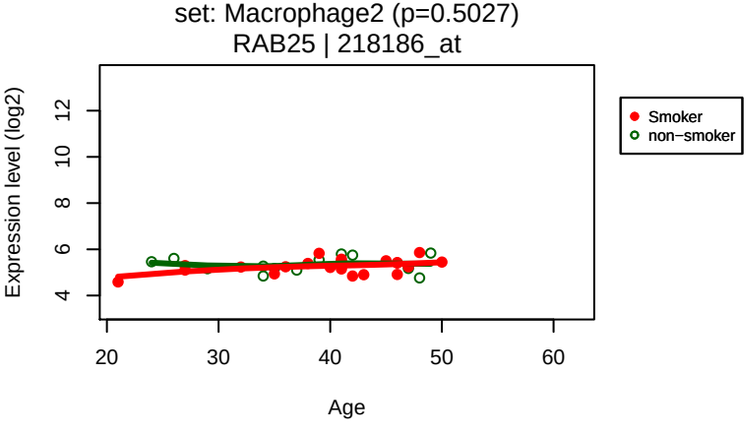

set: SmallAirway2 (p=0.2254)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker

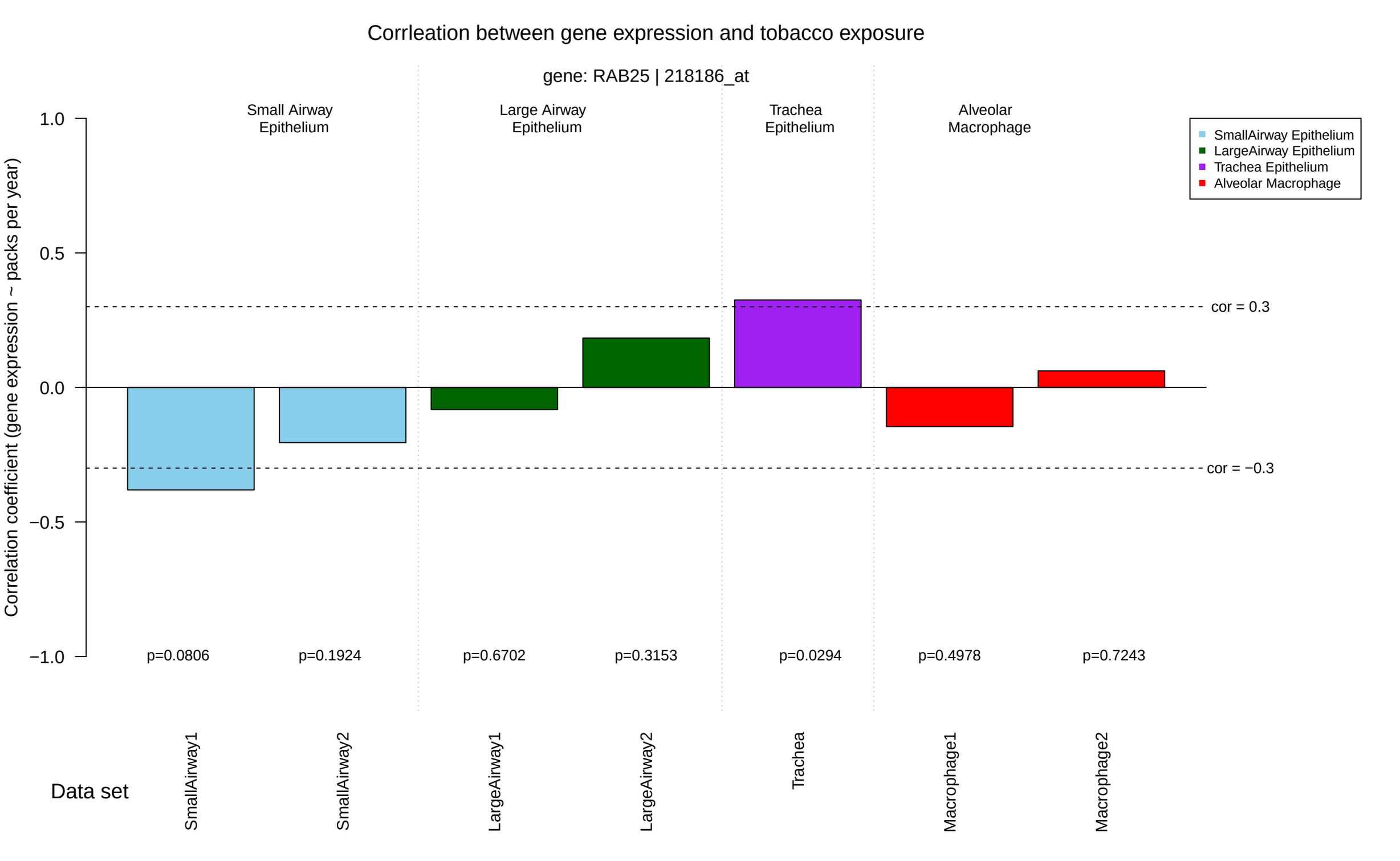

set: LargeAirway1 (p=0.2307)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0563)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0072)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6917)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5027)
RAB25 | 218186_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB25 | 218186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0806
p=0.1924
p=0.6702
p=0.3153
p=0.0294
p=0.4978
p=0.7243
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2